Matrix Theory II–III: Rank and Systems of Linear Equations

Problem sheet for the Day 2 classes · CSIR NET / IIT JAM previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. JAM 2019 NAT If the system $x_1 + x_2 + x_3 + x_4 = 4,\ x_1 + 2x_2 + 3x_3 + 4x_4 = 5,\ x_1 + 3x_2 + 5x_3 + kx_4 = 5$ has no solution, then $k =$ _____.

2. CSIR NET Dec 2016 MCQ Let J denote the matrix of order $n \times n$ with all entries 1 and let B be a $(3n) \times (3n)$ matrix given by $B = \begin{pmatrix} 0 & 0 & J \\ 0 & J & 0 \\ J & 0 & 0 \end{pmatrix}$. Then the rank of B is

(A) $2n$ (B) $3n - 1$
(C) 2 (D) 3

3. CSIR NET Dec 2016 MCQ Let A be an $n \times m$ matrix with each entry equal to $+1, -1$ or 0 such that every column has exactly one $+1$ and exactly one -1. We can conclude that

(A) $\mathrm{Rank}\, A \leq n - 1$ (B) $\mathrm{Rank}\, A = m$
(C) $n \leq m$ (D) $n - 1 \leq m$

4. CSIR NET Dec 2015 MCQ Let A be a real 3×4 matrix of rank 2. Then the rank of A^tA, where A^t denotes the transpose of A, is

(A) exactly 2 (B) exactly 3
(C) exactly 4 (D) at most 2 but not necessarily 2

5. CSIR NET June 2016 MCQ Let A be an $n \times m$ matrix and b be an $n \times 1$ vector (with real entries). Suppose the equation $Ax = b,\ x \in \mathbb{R}^m$ admits a unique solution. Then we can conclude that

(A) $m \geq n$ (B) $n \geq m$
(C) $n = m$ (D) $n > m$

6. CSIR NET June 2019 MCQ What is the rank of the matrix $\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 & 3 \\ 1 & 2 & 3 & 4 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix}$?

(A) 2 (B) 3

(C) 4 (D) 5

7. CSIR NET June 2018 MCQ Let A be an $m \times n$ matrix and B be an $n \times m$ matrix over the real numbers with $m < n$. Then

(A) AB is always nonsingular (B) AB is always singular

(C) BA is always nonsingular (D) BA is always singular

8. CSIR NET Dec 2015 MSQ Let A be an $m \times n$ real matrix and $b \in \mathbb{R}^m$ with $b \neq 0$. Which of the following are true?

(A) The set of all real solutions of $Ax = b$ is a vector space

(B) If u and v are two solutions of $Ax = b$, then $\lambda u + (1-\lambda)v$ is also a solution for any $\lambda \in \mathbb{R}$

(C) For any two solutions u, v, $\lambda u + (1-\lambda)v$ is a solution only when $0 \leq \lambda \leq 1$

(D) If the rank of A is n, then $Ax = b$ has at most one solution

9. CSIR NET Dec 2017 MSQ Let A be an $m \times n$ matrix with rank r. If the linear system $AX = b$ has a solution for each $b \in \mathbb{R}^m$, then

(A) $m = r$

(B) the column space of A is a proper subspace of $\mathbb{R}^m$

(C) the null space of A is a non-trivial subspace of $\mathbb{R}^n$ whenever $m = n$

(D) $m \geq n$ implies $m = n$

10. CSIR NET Dec 2014 MSQ Let A be a 4×7 real matrix and B a 7×4 real matrix such that $AB = I_4$. Which of the following are always true?

(A) $\mathbf{rank}(A) = 4$ (B) $\mathbf{rank}(B) = 7$

(C) $\mathbf{nullity}(B) = 0$ (D) $BA = I_7$